

THE EFFECT OF BATTER REST TIME ON MACARON MERINGUE PROPERTIES

Layal Barakat

MIT Department of Mechanical Engineering
Cambridge, MA, USA

ABSTRACT

Macarons are a French dessert that even expert bakers find hard to perfect. Many different factors such as humidity, ingredient ratios, and batter rest time can affect a macaron's material and geometric properties. The macaron batter must rest before being placed in the oven, and during this time, the macarons form a dried-out surface layer. Macaron batter rest times were varied between 0 to 50 minutes to determine the relationship between rest time, height after baking, change in area, moisture loss, shell hardness, and chewiness. Across all batter rest times from 10 to 50 minutes, the height after baking decreased by 8% and chewiness decreased by 35%. The largest change in area occurred at 30 minutes, with about a 35% decrease in macaron area after baking.

Keywords: Macarons, macaron feet, meringue, chewiness

INTRODUCTION

French macarons, the dessert “sandwich” made with two meringues and a jam-like filling, are one of the most challenging baked goods to make. For such a difficult process, they are made with just a few main ingredients: egg whites, granulated sugar, powdered sugar, and almond flour. A “perfect” macaron meringue is crispy on the outside, chewy on the inside, and develops a bubbly “foot” or “crown” at the bottom edge of the shell. Figure 1 shows what the macaron meringues look like, for reference.

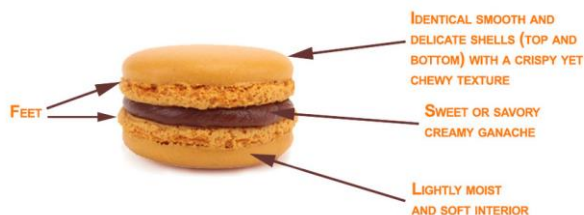


Figure 1: The anatomy of a macaron [3].

There are many reasons why macarons may fail to look or taste like macarons, from the indoor humidity to oven temperature to ingredient ratios [1]. Among these important factors is the length of time that passes after piping (squeezing a pastry bag opened at the bottom tip) the macaron batter onto a tray in individual circles, and before the tray is placed in the oven. In this time, a dried-out surface layer forms on the batter, which allows the macarons to contract, rise, and develop the foot/crown as they bake [1]. This contrasts with more traditional baked goods that primarily spread laterally as they bake, such as cookies. The material and geometric properties of macarons are anecdotally known to be affected by the development of this outer layer, as it directly relates to how the macaron will bake [2].

The relationship between batter rest time and the material and geometric properties of macarons was investigated and quantified to better understand how macarons bake and what variables produced the best macarons. Macarons were baked at a constant temperature and time, while the batter rest time was varied from 0 to 50 minutes in 10-minute intervals. The meringues were tested using a texture analyzer puncture test to measure the shell hardness and to find the height after baking. Other material properties were analyzed as well, such as shell hardness and chewiness. Additionally, image analysis was done to compute change in area. Across all batter rest times from 10 to 50 minutes, the height after baking decreased by 8% and chewiness decreased by 35%. The largest change in area occurred at 30 minutes, with about a 35% decrease in macaron area after baking. There was no statistically significant relationship found between batter rest time, moisture loss, and shell hardness.

BACKGROUND

MACARON INGREDIENTS AND BAKING PROCESS

French macarons are very difficult to bake, and any change to the ingredients or recipe can change the results

drastically [2]. Bakers of all experience levels know that macarons require time and precision to bake a successful batch, and that the ratios of ingredients matter immensely [4]. A typical macaron recipe splits the ingredients into two parts: one, whisking the egg whites and granulated sugar to create stiff peaks for the meringue and two, mixing and sifting the powdered sugar and almond flour. Once each part is complete, the dry ingredients are slowly folded into the meringue mixture, and then piped onto a silicone baking sheet in equidistant and equal sized circles. The tray is “dropped” to flatten the macaron circles and help eliminate any air bubbles, leaving a smooth surface [3]. It is then that the batter is left to rest, and then placed in the oven to bake. The filling is piped onto one macaron meringue and sandwiched by another.

PRIOR STUDIES

An important part of the macaron process is whipping the eggs and sugar into the meringues, and much research has been done on egg white foams [2][4][6]. Whipping breaks down the egg’s proteins and introduces air into the mixture, crucial for the texture of macarons and development of feet. Because egg proteins have hydrophobic (repellant of water) and hydrophilic (attracted to water) ends, the proteins are stretched, with the hydrophobic portions surrounding air bubbles and the hydrophilic portions stretching into the liquid mix [2]. It is important that whipping is not too excessive or else the proteins may stick to each other and reduce the amount of air a water incorporated in the meringue mixture. Excess sugar can also interfere with protein unfolding because it is hygroscopic, meaning it absorbs water [4]. Fats, such as those found in egg yolks, can also interfere with the protein structure because they are hydrophobic and forces air bubbles out, so it is important to only use egg whites. It is also important to note that almond flour is high in fat, so it must be incorporated into the batter only after the meringue is complete and has formed stiff peaks [4].

Another common topic of macaron research is the effect of sugar ratios on a macaron’s properties [2][5]. Macarons use two types of sugar, granulated and powdered. The granulated sugar is added gradually during the egg white whipping process to make the meringues, while the powdered sugar is added later in the process to provide bulk and texture [1]. The distinction between the two sugars is very important, as the powdered sugar dissolves more easily into the macaron batter, creating a lighter texture, while the granules of the granulated sugar provide structure to the meringue [5].

While there is a lot of research on ingredient variations in macarons and the different processes present when whipping meringues, it is not clear that macaron batter rest time has been explicitly varied and its effects studied in an academic setting. Thus, further research is needed to determine how macaron material and geometric properties are affected by batter rest time.

ANECDOTAL EFFECTS OF BATTER REST TIME

There has been anecdotal evidence in the baking community of significant variation in macarons just by changing batter rest time. Macaron batter is viscous enough to hold its shape once it is piped onto a tray, however, it is still very moist. The dried-out surface layer forms as the macaron loses surface moisture to the environment through evaporation. The longer the rest time, the thicker the dried-out surface layer. Too little rest does not allow the dried-out surface layer to properly form. Thus, the macarons develop “spread feet” that spread horizontally instead of vertically. Too much rest means that the macarons have developed a very thick surface layer and the macaron is too dry. This can cause no feet to form at all, as air bubbles in the meringue mixture are not able to escape and form the feet. As a result, this paper seeks to quantify this anecdotal knowledge.

EXPERIMENTAL DESIGN OVERVIEW AND APPARATUS

The shell hardness and chewiness were measured using a texture analyzer (force range 0-50N, 0.01N+/-10N resolution) and the TA-52 puncture probe (2mm diameter, 25mm length). Each macaron was punctured at its center after baking and force-displacement is measured. The heights of the macarons after baking were measured in millimeters (mm) using the texture analyzer. The probe was zeroed on the texture analyzer base plate and the displacement before puncture was averaged to characterize height. Shell hardness (puncture force) and chewiness (pullout) are presented in Newtons (N). Shell hardness is defined as the maximum force at puncture, and chewiness is defined as the magnitude of the minimum negative force during pullout. The macarons were weighed before and after baking on an Ohaus scale (420g maximum, 0.01g resolution) to quantify moisture loss in grams (g). Photos were taken from above the macarons before and after baking. Photoshop and MATLAB were used to process these images for the purpose of calculating change in area by counting pixels in the macaron region.

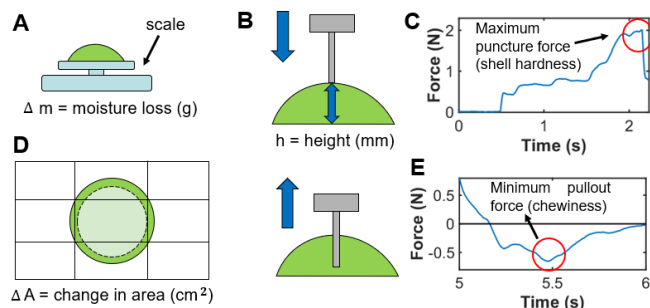


Figure 2: Diagram of full experiment setup. A, a visual of the mass measurement to compute moisture loss. B, a visual of the measure of height by using the texture analyzer. Zeroed at base. C, a visual of texture analyzer puncture probe descending and the resulting plot. Maximum puncture force is shell hardness. D, a visual of the image analysis photos used to find change in area. E, a visual of the puncture probe ascending and the resulting pullout plot. Minimum pullout force is chewiness.

BAKING THE MACARONS

There are two major sets of steps that need to be done precisely. The first is to bake the macarons. The recipe used called for 70 grams of almond flour, 90 grams of powdered sugar, 100 grams of granulated sugar, 30 grams of cocoa powder, and two egg whites. Once the batter was ready, macarons were piped onto silicone sheets directly on the scale to measure the mass before baking. At this point, a photo was also taken from above for each macaron for later use in image analysis. Once six macarons were piped onto six individual silicon sheets, the macarons were left to rest for the specified time between 0 and 50 minutes. The macarons were then baked at 350 degrees Fahrenheit for 12 minutes. They were left on the sheets until cool and stored in airtight containers for one day.

The second major step is to take data on the macarons after one day after baking. First, the mass is measured again using the scale. Then, a picture was taken from above. Data was then taken using the texture analyzer and from this, the shell hardness and chewiness were found.

RESULTS AND DISCUSSION

All macarons were piped from the same batch of batter. Since the batter was relocated to a piping bag, the batter remained moist throughout the process. To avoid overlap in the oven, the order of waiting times and baking

was 0 minutes, 20 minutes, 40 minutes, 30 minutes, 50 minutes, then 10 minutes. The rest time was measured from the moment the last macaron of a six-macaron set was piped. It took between two to three minutes to pipe all six macarons. It is important to note that all of this data is likely exclusive to the recipe used. All error bars in the following plots (figures 3-6) represent the 95% confidence intervals of the six macarons at each batter rest time.

The initial height of the macaron was assumed to be approximately the same across all macarons, so only the height after baking was considered. The data shows that the macaron height decreases with increasing batter rest time. A linear fit was found for the data points from 10 to 50 minutes, shown in figure 3. The zero-minute data was excluded from this fit because it does not have enough time to form a surface layer, while the 10 to 50-minute macarons do. The data shows that as the batter rest time increases, the height of the macaron decreases. The macaron mostly elevates as the feet form and grow at the base of the macaron while baking. One possibility to explain this data is that because the longer batter rest times mean that the macaron is more dried out, the feet were not able to grow as tall. If the anecdotal knowledge is considered (i.e., very long batter rest times cause no feet), then this data provides a trendline that may support that knowledge. Quantitatively, the height after baking decreased by 8% across the 10 to 50-minute batter rest times, which leads to a 2% reduction in height as batter rest time increases by 10 minutes.

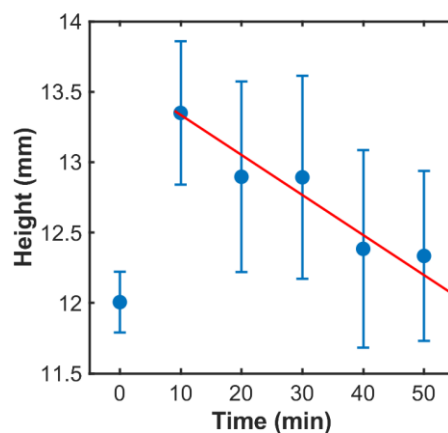


Figure 3: Height of macaron in millimeters versus batter rest time. The fit was of the form $y = Ax + B$ where $A = -0.028 \pm 0.018$ mm/min and $B = 13.62 \pm 0.60$ mm. As batter rest time increases by 10 minutes, the macaron height decreases by 2%. An overall decrease of 8% was found across 10-50 minutes. The data

point at the 0-minute batter rest time was excluded from the fit as it was an outlier.

Selecting more batter rest times, especially upwards of an hour could prove to be interesting, as it is likely that a plateau might be reached in the data. Since there is a downward trend, it could be possible to find the batter rest time which provides the shortest macaron. Otherwise, the tallest macarons are achieved with a batter rest time of 10 minutes. Bakers have different preferences for the ideal macaron height, but this data shows that there is a maximum height reached at 10 minutes. Bakers who prefer shorter macarons can still use this data to fine tune the height they would like to achieve.

The shell hardness and chewiness were both found using the texture analyzer. Maximum peak force at puncture was the shell hardness, and the magnitude of the minimum pullout force was the chewiness. The two respective plots versus batter rest time are found in figure 4. The two plots versus batter rest time can be found in figure 4. While there was no statistically significant relationship between shell hardness and batter rest time, a linear fit was found for the chewiness data. The zero-minute data was again excluded from the fit because of the lack of surface layer. While the 10 to 50-minute macarons have a dried-out surface layer, the 0-minute macarons do not. The trendline shows that as batter rest time increases by 10 minutes, the chewiness decreases by about 7%, with a 35% total decrease across the 10 to 50-minute batter rest times. This can be explained like the negative height trend in the previous graph. It is likely that with increased rest times, there is decreased moisture in the macaron, which leads to less chewiness. Most bakers prefer a chewier macaron, and this data shows the maximum chewiness was found at the 10-minute batter rest time.

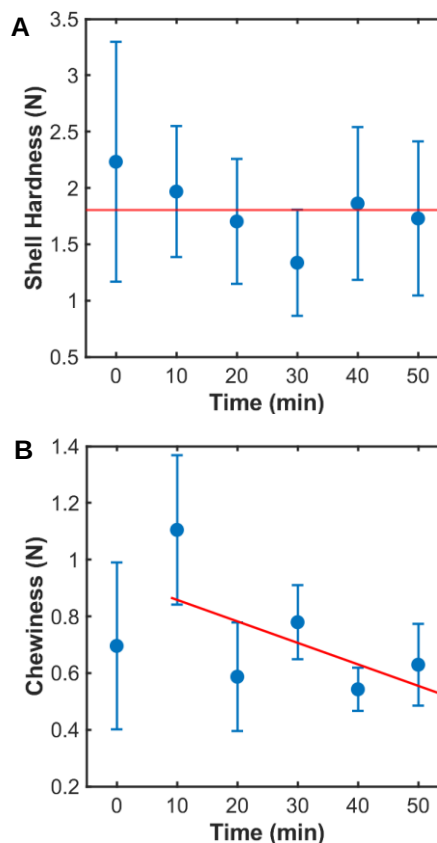


Figure 4: Texture analyzer plots. A, shell hardness (maximum puncture force) versus batter rest time is on the right. B, chewiness (magnitude of minimum pullout force) versus batter rest time is on the right. Shell hardness data was not statistically significant, as shown by the horizontal line. Chewiness was statistically significant, with a fit of the form $y = Ax + B$, where $A = -0.008 \pm 0.006$ N/min and $B = 0.93 \pm 0.20$ N. As a general trend, as batter rest time increases by 10 minutes, chewiness decreases by about 7%. There was a 35% decrease in chewiness across 10-50 minutes. Maximum chewiness is achieved with a 10-minute batter rest time.

The macarons' mass at piping and a day after baking was measured, and the difference was defined as the moisture loss. To gain an understanding of the data, the moisture loss in grams was divided by the initial mass of that macaron and multiplied by 100 to get a percent loss. The plot of moisture loss versus batter rest time can be found in figure 5. There was no statistically significant relationship found between moisture loss and batter rest

time, which is at odds with the previous theory that chewiness and height decrease with decreasing moisture.

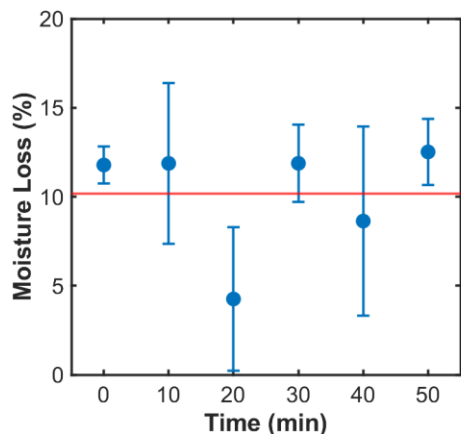


Figure 5: Percent moisture loss versus batter rest time. Relationship is not statistically significant, as shown by the horizontal line. This is at odds with the theories proposed for previous fits.

One potential reason for this is that this total moisture loss calculated here is the sum of moisture lost to the environment plus moisture lost in the oven. The moisture lost in the oven is unknown. It could be constant or proportional based on the starting amount of moisture. It is still possible that the moisture lost to the environment was linearly related to the batter rest time. Future iterations of this project could measure the weight of the macaron before it is placed in the oven to differentiate between these different sources of moisture loss.

One more thing to consider is that the humidity or weather that day could have also played a role in how much moisture was lost to the environment. Macarons dry more in a less humid environment than a more humid one. The more moisture that is in the air, the slower the macarons would form the surface layer. Though the humidity was not measured in this experiment, it would be an interesting extension to a future experiment.

The last important parameter was the change in area. Photos taken before and after baking reveal that all macarons decreased in size laterally. To quantify the data more clearly, the change in area (in cm^2) was divided by the initial area and multiplied by 100 to get a percent change in area. The plot of change in area versus batter rest time is found in figure 6. In contrast to all previous graphs which either had no statistically significant relationship or had a linear relationship, the fit of this data is of a second-degree polynomial. This again excludes the zero-minute data from the fit, for the same reasons as

mentioned for other fits: there was not enough time for a surface layer to form.

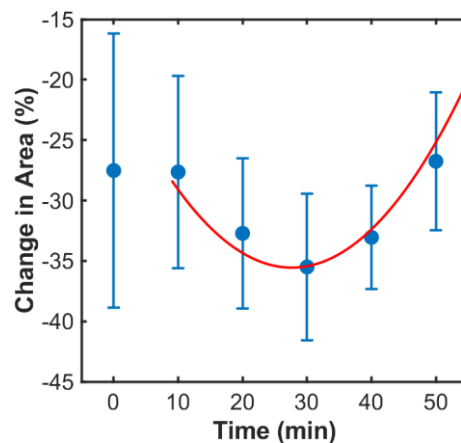


Figure 6: Percent change in area versus batter rest time. Relationship is statistically significant with a fit of the form $y = Ax^2 + Bx + C$, where $A = 0.021 \pm 0.013 \text{ min}^{-1}$, $B = -1.15 \pm 0.81 \text{ N}$, and $C = -20. \pm 11 \text{ N}$. The zero-minute data point was not included in the fitting. The minimum of the parabola represents the greatest change in area at around the 30-minute batter rest time mark, with a -35% change in the area (shrink).

As also mentioned for the height data, further research could expand upon the batter rest times, seeing if and when there is a plateau in the data after 50 minutes of rest. This could help to find what batter rest time results in the smallest possible area change, if that is something desired. Otherwise, the maximum possible area change can be found on this graph. The minimum of the fit parabola represents the greatest change in area at around the 30-minute batter rest time mark, with a 35% decrease in the area. This means that the final macarons that rested for 30 minutes were about 35% smaller than their pre-baking counterparts. This result is quite different than the linear relationship that was expected.

CONCLUSIONS

Across all batter rest times from 10 to 50 minutes, the height after baking decreased by 8% and chewiness decreased by 35%. The largest change in area occurred at 30 minutes, with about a 35% decrease in macaron area after baking. The shell hardness and moisture loss do not seem to be affected by different batter rest times. The smallest macarons by area occurred at the 30-minute batter rest time. The 10-minute batter rest time resulted in the tallest and chewiest macarons, making it the ideal

for bakers who prefer those qualities. Even for those who do not, the data can be used to optimize for different properties and achieve specific qualities in their macarons.

Future iterations of this experiment could use different macaron recipes to see what effect ingredient ratios have on the surface layer. Other iterations could try measuring or varying the humidity to study the drying of the surface layer in depth, or measuring the macaron's moisture at different points of the moisture loss process. Selecting more batter rest times, especially upwards of an hour could prove to be interesting, as it is likely that a plateau might be reached in collected data. This plateau could be helpful in achieving shorter macarons or macarons with the smallest change in area. It could be useful to select batter rest times that are more spread out to also try and reach this plateau. Additionally, this extension would likely also decrease the error found due to the earlier piped macarons in every set of six having slightly more time to dry.

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